Q1. (a) Find the point at which the given lines intersect: **[5pts]**

$$L_1: \; x = t,\; y = 2 - t,\; z = -2 + 2t$$
$$L_2: \; x = 3 - s,\; y = -1 + s,\; z = -2 + s$$

We need to solve the following system:

$$\begin{cases} t = 3 - s & (1) \\ 2 - t = -1 + s & (2) \\ -2 + 2t = -2 + s & (3) \end{cases}$$

If we substitute $t = 3 - s$ in (3), we obtain: $4 - 2s = -2 + s$ i.e $s = 2$.

Thus the coordinates of the intersection point are:

$$x = 1,\; y = 1,\; z = 0$$

Rmk: It is easy to see that when $s = 2$, $t = 1$. If we substitute $t = 1$ in the parametric equations of L_1, we find: $x = 1$, $y = 1$ and $z = 0$.

(b) Find an equation of the plane that contains the lines L_1 and L_2. **[7pts]**

$V_1 = \langle 1, -1, 2 \rangle$ is parallel to L_1
$V_2 = \langle -1, 1, 1 \rangle$ is parallel to L_2
} $V_1 \times V_2$ is orthogonal to the plane that contains L_1 and L_2.

$$V_1 \times V_2 = \begin{vmatrix} i & j & k \\ 1 & -1 & 2 \\ -1 & 1 & 1 \end{vmatrix} = -i - 2j + k - k - 2i - j = -3i - 3j = -3\langle 1, 1, 0 \rangle$$

Hence the vector $n = \langle 1, 1, 0 \rangle$ is normal to the plane.

An equation of the plane is given by:

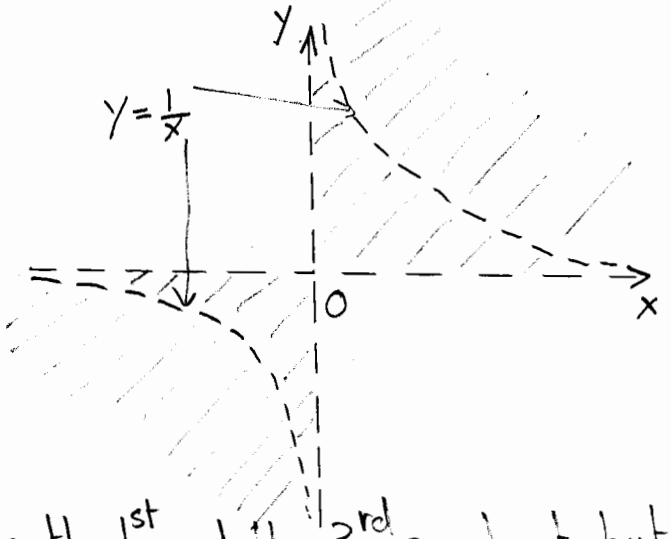
$$(x - 1) + (y - 1) = 0 \quad \text{or} \quad x + y - 2 = 0$$

Q2. Find and sketch the domain of the function $f(x,y) = \dfrac{1}{\ln(xy)}$. **[5pts]**

$f(x,y)$ is well defined iff $xy>0$ and $\ln(xy)\neq 0$

$xy>0$ iff "$x>0$ and $y>0$" or "$x<0$ and $y<0$"

$\ln(xy)\neq 0$ iff $xy\neq 1$ iff $y\neq \frac{1}{x}$



The domain of f is the 1st and the 3rd quadrant, but the x-axis, the y-axis and the graph $y=\frac{1}{x}$ are not included.

Q3. Find the linear approximation $L(x,y)$ of the function $f(x,y) = x^2y + \sqrt{x^2+y^2}$ at the point $(1,0)$ and use it to approximate the value of $f(0.98, 0.03)$. **[10pts]**

$$f(x,y) \approx f(1,0) + f_x(1,0)(x-1) + f_y(1,0)\,y$$

$f_x(x,y) = 2xy + \dfrac{x}{\sqrt{x^2+y^2}}$. Thus $f_x(1,0) = 1$

$f_y(x,y) = x^2 + \dfrac{y}{\sqrt{x^2+y^2}}$. Thus $f_y(1,0) = 1$

Hence: $f(x,y) \approx 1 + x - 1 + y$

$\approx x + y$

Therefore $f(0.98, 0.03) \approx 1.01$

Q4. (a) Let $z = \tan^{-1}\left(\frac{x}{y^2}\right)$. Find $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$ at the point $(x, y) = (-1, -1)$. **[5pts]**

$$\frac{\partial z}{\partial x} = \frac{\frac{1}{y^2}}{1 + \left(\frac{x}{y^2}\right)^2}. \text{ Thus } \left.\frac{\partial z}{\partial x}\right|_{(-1,-1)} = \frac{1}{2}$$

$$\frac{\partial z}{\partial y} = \frac{-\frac{2x}{y^3}}{1 + \left(\frac{x}{y^2}\right)^2}. \text{ Thus } \left.\frac{\partial z}{\partial y}\right|_{(-1,-1)} = -1$$

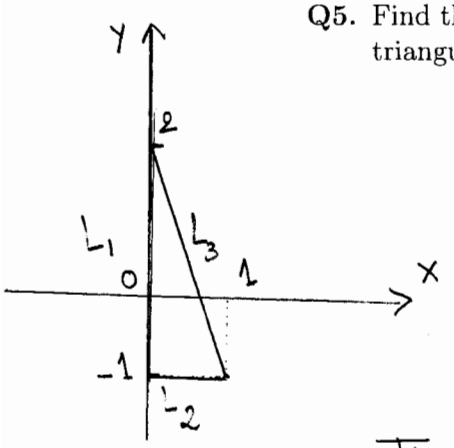
Hence: $\left.\frac{\partial z}{\partial x}\right|_{(-1,-1)} + \left.\frac{\partial z}{\partial y}\right|_{(-1,-1)} = -\frac{1}{2}$

(b) Let $F(x, y) = x\cos(y) + \sin(xy)$. Find $F_{xyx}(1, \pi/2)$. **[6pts]**

- $F_x(x,y) = \cos y + y\cos(xy)$
- $F_{xy}(x,y) = -\sin y + \cos(xy) - xy\sin(xy)$
- $F_{xyx}(x,y) = -y\sin(xy) - y\sin(xy) - xy^2\cos(xy)$
  $= -2y\sin(xy) - xy^2\cos(xy)$

Hence: $F_{xyx}\left(1, \frac{\pi}{2}\right) = -\pi$

Q5. Find the absolute maximum and minimum values of $f(x,y) = e^{x^2y}$ on the closed triangular region with vertices $(0,-1)$, $(0,2)$ and $(1,-1)$. **[13pts]**



• $f_x(x,y) = 2xy e^{x^2y}$, $f_y(x,y) = x^2 e^{x^2y}$

Setting $f_x(x,y) = 0 = f_y(x,y)$ implies:

$$\begin{cases} 2xy\, e^{x^2y} = 0 \\ \& \\ x^2 e^{x^2y} = 0 \end{cases} \Rightarrow \begin{cases} xy = 0 \\ \& \\ x^2 = 0 \end{cases} \Rightarrow \begin{cases} xy = 0 \\ \& \\ x = 0 \end{cases}$$

Thus any point of the form $(0,y)$ with $-1 \le y \le 2$ is a critical point of f. In other words all the points of L_1 are critical points but none of them is in the interior of D.

$f(0,y) = e^0 = 1$

• On L_2: $y = -1$ and $0 \le x \le 1$.

$f(x,-1) = e^{-x^2} = g(x)$. $g'(x) = -2x e^{-x^2}$

x	0		1
g'(x)		−	
g(x)	1	↘	$\frac{1}{e}$

The maximum value of f on L_2 is 0, and it occurs at the point $(0,-1)$

The minimum value of f on L_2 is $\frac{1}{e}$, and it occurs at the point $(1,-1)$.

• On L_3: $y = 2-3x$ and $0 \le x \le 1$

$f(x, 2-3x) = e^{x^2(2-3x)} = e^{2x^2-3x^3} = h(x)$

$h'(x) = (4x - 9x^2) e^{x^2(2-3x)} = x(4-9x) e^{x^2(2-3x)}$

x	0		4/9		1
h'(x)		+	0	−	
h(x)	1	↗	$e^{32/243}$	↘	$\frac{1}{e}$

Thus the maximum value of f on L_3 is $e^{32/243}$, and it occurs at $(\frac{4}{9}, \frac{2}{3})$

The minimum value of f on L_3 is $\frac{1}{e}$, and it occurs at $(1,-1)$

Ans. The abs. max. is $f(\frac{4}{9}, \frac{2}{3}) = e^{32/243}$

The abs. min is $f(1,-1) = \frac{1}{e}$

Q1. Symmetric equations for the line of intersection of the planes $z = 2x - y - 5$ and $z = x + 3y - 1$ are

(A) $\dfrac{x}{-2} = y + 2 = \dfrac{z+3}{7}$

(B) $\dfrac{x-4}{4} = -y = \dfrac{z-3}{7}$

(C) $\dfrac{x-4}{-2} = y = \dfrac{z-3}{7}$

(D) $\dfrac{x}{4} = y + 2 = \dfrac{z+3}{7}$

(E) $\dfrac{x-4}{4} = y = \dfrac{z-3}{7}$

$x + 3y - 1 = 2x - y - 5$

$\Rightarrow 4y = x - 4 \Rightarrow y = \dfrac{x-4}{4}$.

Thus: $z = (4y + 4) + 3y - 1$

$\Rightarrow y = \dfrac{z-3}{7}$.

Q2. Let $W(s,t) = F(u(s,t), v(s,t))$, where F, u and v are differentiable functions and

$$u(1,0) = 2, \quad v(1,0) = -3, \quad u_s(1,0) = -2, \quad v_s(1,0) = 5,$$
$$u_t(1,0) = -6, \quad v_t(1,0) = 4, \quad F_u(2,-3) = 1, \quad F_v(2,-3) = 3.$$

The value of $W_s(1,0) + W_t(1,0)$ is equal to

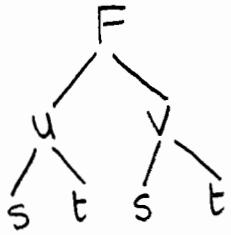
(A) 22

(B) 18

(C) 19

(D) 21

(E) 20



$\cdot) W_s(s,t) = \dfrac{\partial F}{\partial u} \cdot \dfrac{\partial u}{\partial s} + \dfrac{\partial F}{\partial v} \cdot \dfrac{\partial v}{\partial s}$

Thus $W_s(1,0) = F_u(u(1,0), v(1,0))\, u_s(1,0)$
$+ F_v(u(1,0), v(1,0))\, v_s(1,0)$

$= F_u(2,-3)\, u_s(1,0) + F_v(2,-3)\, v_s(1,0)$

$= -2 + 15 = 13$

$\cdot) W_t(s,t) = \dfrac{\partial F}{\partial u} \cdot \dfrac{\partial u}{\partial t} + \dfrac{\partial F}{\partial v} \cdot \dfrac{\partial v}{\partial t}$. Thus:

$W_t(1,0) = F_u(2,-3)\, u_t(1,0) + F_v(2,-3)\, v_t(1,0)$

$= -6 + 12 = 6$

Hence: $W_s(1,0) + W_t(1,0) = 19$

Q3. $\lim\limits_{(x,y)\to(0,0)} \dfrac{1-\cos(x^2+y^2)}{\sin^2(2x^2+2y^2)}$ is equal to

(A) $\frac{1}{4}$

(B) 2

(C) 1

(D) $\frac{1}{8}$

(E) $\frac{1}{2}$

$$\sin^2(2x^2+2y^2) = 4\sin^2(x^2+y^2)\cos^2(x^2+y^2)$$
$$= 4\left[1-\cos^2(x^2+y^2)\right]\cos^2(x^2+y^2)$$
$$= 4\left[1+\cos(x^2+y^2)\right]\left[1-\cos(x^2+y^2)\right]\cos^2(x^2+y^2)$$

Thus: $$\lim_{(x,y)\to(0,0)} \frac{1-\cos(x^2+y^2)}{\sin^2(2x^2+2y^2)} = \lim_{(x,y)\to(0,0)} \frac{1}{4\left[1+\cos(x^2+y^2)\right]\cos^2(x^2+y^2)}$$
$$= \frac{1}{8}$$

Q4. Let $T(x,y) = e^{x^2-y^2}$. The function T increases most rapidly at the point $(-1,-1)$ in the direction of the vector

(A) $\langle -1, 1\rangle$

(B) $\langle 1, -2\rangle$

(C) $\langle -2, 1\rangle$

(D) $\langle 1, -1\rangle$

(E) $\langle -1, -1\rangle$

$$\nabla T(x,y) = \langle 2x e^{x^2-y^2}, -2y e^{x^2-y^2}\rangle$$

Thus $\nabla T(-1,-1) = \langle -2, 2\rangle = 2\langle -1, 1\rangle$

Q5. The equation $-2x^2 - y^2 + z^2 - 2y - 4z - 12x - 15 = 0$ represents

(A) a hyperboloid of one sheet

(B) a cone

(C) a hyperboloid of two sheets

(D) an ellipsoid

(E) a hyperbolic paraboloid

$$-2(x^2+6x)-(y^2+2y)+(z^2-4z)-15=0$$

$$\Rightarrow -2\left[(x+3)^2-9\right]-\left[(y+1)^2-1\right]+\left[(z-2)^2-4\right]-15=0$$

$$\Rightarrow -2(x+3)^2+18-(y+1)^2+1+(z-2)^2-4-15=0$$

$$\Rightarrow -2(x+3)^2-(y+1)^2+(z-2)^2=0$$

$$\Rightarrow (x+3)^2+\frac{(y+1)^2}{(\sqrt{2})^2}=\frac{(z-2)^2}{(\sqrt{2})^2}$$

Which represents a cone

Q6. Consider the surface $\overbrace{x^2z + 3yz^2 + 3xyz}^{F(x,y,z)} = 7.$

Let $Ax + By + 10z = D$ be an equation of the tangent plane to the given surface at $(1, 1, 1)$. The value of $A + B + D$ is equal to

(A) 39

(B) 37

(C) 42

(D) 32

(E) 36

$$\nabla F(x,y,z)=\langle 2xz+3yz,\ 3z^2+3xz,\ x^2+6yz+3xy\rangle$$

Thus $\nabla F(1,1,1)=\langle 5,6,10\rangle$

An equation of the tangent plane is:

$$5(x-1)+6(y-1)+10(z-1)=0$$

or $5x+6y+10z=21$. Therefore:

$A=5$, $B=6$, $D=21$. Hence

$$A+B+D=32.$$

Q7. The function $f(x,y) = 2x^3y + 3x^2 + y^2$ has

(A) three saddle points

(B) one local minimum and two saddle points

(C) two local minima and one local maximum

(D) one saddle point and two local maxima

(E) one local maximum and two saddle points

•) $f_x(x,y) = 6x^2y + 6x = 6x(xy+1)$, $f_y(x,y) = 2x^3 + 2y$

Setting $f_x(x,y) = 0 = f_y(x,y)$ implies: $\begin{cases} 6x(xy+1) = 0 \\ \text{and} \\ 2x^3 + 2y = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \text{ or } xy = -1 \quad (1) \\ \& \\ 2x^3 + 2y = 0. \quad (2) \end{cases}$

Now: If $x = 2$, (2) implies $y = 0$. Thus $(0,0)$ is a critical point.

If $xy = -1$ i.e $y = -\frac{1}{x}$, (2) implies: $2x^3 - \frac{2}{x} = 0$ i.e $x^4 = 1$

i.e $x = \pm 1$. Then $(1,-1)$ and $(-1,1)$ are critical points

•) $f_{xx}(x,y) = 12xy + 6$, $f_{yy}(x,y) = 2$ and $f_{xy}(x,y) = 6x^2$

•) $D(0,0) = 12 > 0$. Since $f_{xx}(0,0) = 6 > 0$, $(0,0)$ is a local min.

•) $D(1,-1) = (-12)(2) - 36 < 0$. Thus $(1,-1)$ is a saddle point.

•) $D(-1,1) = (-12)(2) - 36 < 0$ Hence $(1,-1)$ is saddle point.